Softmax Inundation and Generational Context in Long-Horizon Agent Systems

Analytical Attention Dilution Bounds and the Generational Memory Architecture

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Abstract. *Autonomous AI agent architectures routinely treat the transformer context window as an unbounded, append-only linear interaction transcript. We analyze the theoretical and empirical consequences of this paradigm. We formalize the* ***Softmax Inundation Bound****, an analytical upper bound showing that under bounded projection energy, the maximum aggregate self-attention probability mass allocated to a fixed set of invariant constraint tokens decays asymptotically as O(L_u / |S_noise|) when unpruned execution observations accumulate. This bound is conditional: it is informative only while the attention score dynamic range remains moderate, and it is vacuous for large score contrast. Where it does bite, it identifies token volume as the variable a system designer can control. To address volume, we specify* ***Generational Context****, an operating-system-inspired memory architecture that separates memory into* ***Gen-0 Ephemeral*** *(tool outputs distilled to 3-line signatures and evicted in 1 turn),* ***Gen-1 Evaporative*** *(working memory paged via the Attention-Weighted Eviction algorithm), and* ***Gen-2 Tenured*** *(permanently pinned invariants). We prove that Generational Context maintains a non-zero lower bound on invariant attention uniformly for all interaction turns $t \in N$, removing horizon-dependence from the worst case. This is a guarantee about attention allocation, not about task success; live-model validation is future work.*

1. Introduction

Despite expansions in foundation model context windows, practical autonomy in multi-turn software engineering loops remains fragile. Agents frequently lose adherence to system constraints, repeat failed tool invocations, and apply superficial micro-patches around symptoms rather than addressing root causes.

In typical agent sessions, over 90% of active tokens consist of raw tool outputs (compiler stderr, test dumps, DOM trees). We analyze this pathology via **Payload Inundation** (attention dilution), **Error Gravitation** (semantic bias toward error tracebacks), and **Generational Context** (multi-tiered memory management).

2. The Softmax Inundation Bound

Let context tokens be partitioned into Invariant Subspace S_inv (|S_inv| = L_u) and Observation Subspace S_noise. Under bounded query and key energies ||q_i||_2 ≤ M_Q and ||k_j||_2 ≤ M_K, let κ = exp(2 M_Q M_K / sqrt(d_k)).

Theorem 1 (Softmax Inundation Upper Bound).
For any query position i, the aggregate attention probability mass allocated to the invariant subspace S_inv is upper-bounded by:

$$A_{i,inv} \le \kappa \frac{L_u}{L_u + |S_{noise}|} = O\left(\frac{L_u}{|S_{noise}|}\right)$$

Proof Sketch: By Cauchy-Schwarz, -M_Q M_K ≤ q_i^T k_j ≤ M_Q M_K. The numerator is bounded above by L_u exp(M_Q M_K / sqrt(d_k)) and the denominator is bounded below by (L_u + |S_noise|) exp(-M_Q M_K / sqrt(d_k)). Dividing numerator by denominator yields the result directly.

3. Generational Context & Retention Guarantee

Generational Context stratifies memory into three lifecycle tiers: **Gen-0 Ephemeral** (tool outputs distilled to 3 lines and purged in 1 turn), **Gen-1 Evaporative** (working memory paged via AWE), and **Gen-2 Tenured** (permanently pinned invariants).

Theorem 2 (Generational Retention Guarantee).
Under Generational Context with bounded distillation $||\sigma_distill(G_0)|| \le K_0$, working capacity $||G_1|| \le C_1$, and bounded framing overhead C_overhead (delimiters, role tags, headers), the active non-invariant tokens satisfy $|S_non\text{-}inv^(t)| \le C_1 + K_0 + C_overhead = C_bound$ for all turns t. Under uniform bounded projection energy:

$$A_{i, inv} = 1 / (1 + R_t) \ge L_u / (L_u + \kappa C_{bound}) > 0$$

where $R_t \le (\kappa C_bound) / L_u$. Invariant attention remains strictly bounded away from zero uniformly for all interaction turns $t \in N$.

Numerical Sensitivity Analysis & Comparative Bound Analysis (L_u = 100, C_bound = 150)

Regime	Logit Gap (Δs)	Contrast (κ)	GenContext Min Attn	Flat Context (N=5k)	Bound Ratio R(5k)
Illustrative Low Contrast	1.0	2.72	19.70%	0.73%	27.0x
Illustrative Intermediate	2.0	7.39	8.28%	0.27%	30.7x
Illustrative Moderate	3.0	20.09	3.21%	0.10%	32.3x
Illustrative High Contrast	4.0	54.60	1.21%	0.037%	32.9x
Illustrative Extreme Gap	6.0	403.4	0.165%	0.0049%	33.3x

Comparative Bound Analysis: Flat context values represent theoretical lower bounds under identical uniform score assumptions, not empirical observations. The theoretical bound ratio $R(\kappa, N) = (L_u + \kappa N) / (L_u + \kappa C_bound)$ approaches N / C_bound as $\kappa \to \infty$ (33.3x at N = 5,000; 333.3x at N = 50,000). For finite κ, R ranges from 27.0x to 33.3x (at N = 5k) and 268.1x to 332.8x (at N = 50k). This compares theoretical bound expressions, not measured attention distributions in a neural model.

4. The Attention-Weighted Eviction (AWE) Algorithm

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Algorithm 1: AWE with Hard Capacity Invariant
  1: Input: Capacity C, incoming chunk E, boundary parameter p
  2: if E.tokens > C then Truncate E.content to C tokens; E.tokens ← C
  3: Compute IDS(E) = AttentionWeight(E) × 100 / (Entropy(E) + 10^-4) [or tier prior if unmeasured]
  4: if E.key ∈ T1 ∪ T2 then Update and promote to T2
  5: else if E.key ∈ B1 then Adapt p ← min(C, p + |B2|/|B1| × E.tokens); Evict lowest IDS victim
  6: else if E.key ∈ B2 then Adapt p ← max(0, p - |B1|/|B2| × E.tokens); Evict lowest IDS victim
  7: else Evict lowest IDS chunk; Insert E into T1; assert total_tokens() ≤ C
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5. Empirical Token Compaction & Attention Simulation

*Methodological Scope Notice: Attention metrics represent a deterministic 128-dimensional subword hash-projection surrogate modeling softmax dilution mechanics, distinct from neural transformer forward passes. Controlled empirical evaluations with live neural models (e.g., Claude 3.5 Sonnet, GPT-4o, Llama-3-70B) across SWE-bench Verified are **planned future work** (Phase 2); no live model reasoning claims are made prior to Phase 2 completion.*

Architecture Arm	Mean Active Tokens	Context Signal Ratio (CSR)	Simulated Attn(Inv) (Proxy)	Token Reduc
Arm A: Flat Append-Only	5,099	0.0059	0.0425	Baseline
Arm B: Auto-Compaction	3,340	0.0077	0.0553	-34.5%
Arm C: Generational Context	129	0.2000	0.0757	**-97.5% (33.9x CSR)**

6. Related Work & Conclusion

Our work connects with Liu et al. (2023) on context retrieval degradation, Xiao et al. (2023) on attention sinks, Packer et al. (2023) on MemGPT tiered memory, and Megiddo & Modha (2003) on ARC. By formalizing the Softmax Inundation Bound and providing the Generational Context architecture, we offer a deterministic, verifiable memory framework that preserves foundational constraints across arbitrary interaction horizons.

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